

Overview of UNIX/Linux Operating System

MODULE 1: GETTING STARTED & NAVIGATING THE SYSTEM

INDEX

- What is an Operating System?
- What is Unix?
- History of UNIX Operating System
- Linux Operating System
- Features of Linux Operating System
- Structure of LINUX Operating System.
- Kernel in LINUX.
- Shell in LINUX.
- Types of Shell
- Distributions of Linux

What is an Operating System?

- An operating system or OS is software installed on a computer's hard drive that enables the computer hardware to communicate and operate with the computer software.
- In normal word, we can say that an operating system is a messenger which **helps the user to communicate with the hardware.**
- This picture shows latest Microsoft Windows 11 in its original packaging.



What is Unix?

- UNIX is a **multi-user**, **multitasking** operating system (OS) that was developed at **AT&T Bell Labs** in the late 1960s. It was designed to be used exclusively by programmers and became a leading operating system for workstations because of its **portability**, flexibility and power.
- UNIX systems also have a graphical user interface (**GUI**) similar to Microsoft Windows which provides an easy to use environment
- The UNIX operating system is made up of three parts:
  - The Kernel
  - The Shell
  - The Programs

History of UNIX Operating System.

- The UNIX operating system was created in 1969 at by Ken Thompson, Dennis Ritchie, and a team of researchers looking for a simpler, portable alternative to the bulky Multics project.
- The history of UNIX traces its evolution from a simple research project at Bell Labs to one of the most influential operating systems in the world.

  - Origins and PDP-7 (1969)
  - Rewriting in C and Portability (1973)
  - Academic Expansion and Commercialization (1970s – 1980s)
  - Legacy and Modern Influence (1990s – Present)

Origins and PDP-7 (1969)

- **Multics Failure:** Bell Labs pulled out of the overly complex Multics experimental time-sharing project in 1969.
- **The First Version:** Ken Thompson used an idle PDP-7 minicomputer to write a stripped-down file system and kernel.
- **The Name:** Brian Kernighan coined the name "UNIX" as a pun on Multics (Uniplexed Information and Computing Service).

Rewriting in C and Portability (1973)

- **Assembly to C:** Dennis Ritchie created the C programming language, and in 1973, Thompson and Ritchie rewrote the Unix kernel in C.
- **Hardware Independence:** This rewrite made Unix portable across completely different computer hardware architectures rather than being locked to one machine.
- **The Name:** Brian Kernighan coined the name "UNIX" as a pun on Multics (Uniplexed Information and Computing Service).

Academic Expansion and Commercialization (1970s – 1980s)

- **BSD Fork:** Ken Thompson taught Unix at UC Berkeley, leading to the creation of the Berkeley Software Distribution (BSD) branch.
- **System V:** AT&T released commercial versions like UNIX System V, establishing two major rival families (BSD and System V).
- **Commercial Variants:** Tech giants built proprietary versions including Solaris (Sun), HP-UX (HP), and AIX (IBM).

Legacy and Modern Influence (1990s – Present)

- **Birth of Linux:** Frustrated by proprietary licensing, Linus Torvalds created Linux in 1991 as a free, Unix-like alternative.
- **Modern OS Foundation:** Unix design principles directly shape modern operating systems like macOS, iOS, Android, and various Linux distributions.
- It is a picture of Bell Labs (KEN THOMPSON is sitting on the chair and DENNIS RITCHIE is standing next to Ken).



Linux Operating System

- Linux is a **free, open-source operating system** that manages computer hardware and software.
- It was developed by Linus Torvalds in **1991**, Linux officially transitioned to a true open-source model in **1992**. The transition occurred when Linus Torvalds changed the project's legal license, which transformed it from a restrictive student hobby into a globally collaborative phenomenon.
- Now-a-days from smartphones to cars, supercomputers and home appliances, home desktops to enterprise servers, the Linux OS is everywhere.

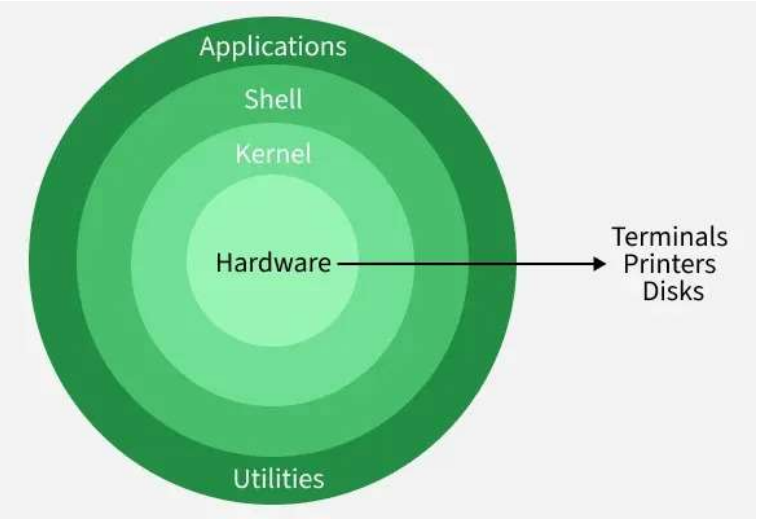
Features of Linux Operating System

- **Open Source:** The source code is freely available to everyone. You can view, modify, and distribute it.
- **Security:** Linux is very secure. It uses user privilege controls, password protection, and encrypted files. It rarely gets viruses compared to other systems.
- **Multi-user:** Multiple users can use the same system at the same time. They can run their own programs without disturbing each other.
- **Multitasking:** Linux can run many programs at the same time without slowing down.
- **Portability:** Linux works on many different hardware devices. You can run it on small computers, large servers, and mobile phones. Security
- **Customizability:** You can change almost everything in Linux. You can pick different desktop looks, themes, and system tools.
- **Stability:** Linux rarely crashes. It can run for years without needing a restart. This makes it great for web servers.
- **Command Line and GUI:** Linux offers a powerful text command line for experts. It also offers easy-to-use graphic desktop screens for regular users.

Structure of Linux Operating System

The Linux system basically works on **4 layers**.

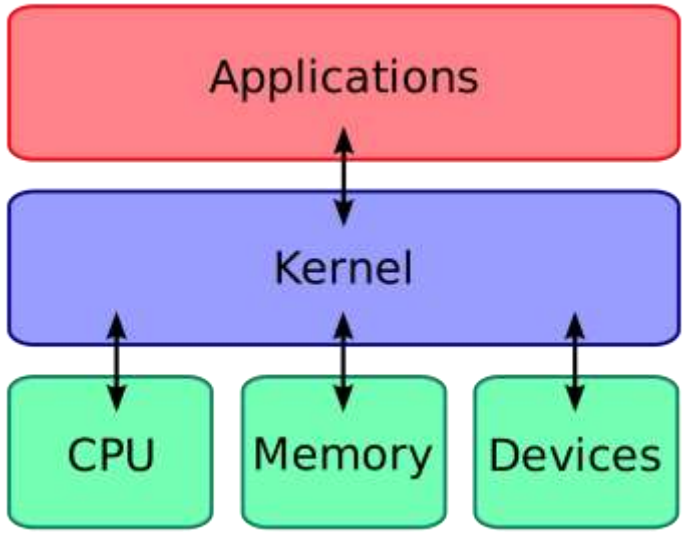
- **Hardware** – Hardware consists of all physical devices attached to System. For example: Hard disk drive, RAM, Motherboard, CPU etc.
- **Kernel** – Kernel is the core component for any (Linux) operating system which directly interacts with the hardware.
- **Shell** – Shell is the interface which takes input from Users and sends instructions to the Kernel.
- **Applications** - These are the utility programs which runs on Shell.



- **System Utilities** - System Utilities are built-in tools used for system management, configuration, and maintenance.

Kernel in Linux

- The **Linux kernel** is the absolute core component of the Linux operating system. It acts as a **critical bridge and translator** between the computer's physical hardware and the software applications running on it.



Shell in Linux

In Linux, a **shell** is a command-line interpreter that acts as an interface between the user and the operating system kernel.

The Core Hierarchy

- **Terminal:** The window or text-based program where you type (e.g., GNOME Terminal).
- **Shell:** The program running *inside* the terminal that reads, interprets, and manages your commands.
- **Kernel:** The core software that directly interacts with your computer's physical hardware.

Types of Shell

- Bourne shell (sh)
- Korn shell (ksh)
- Bourne Again shell (bash)
- POSIX shell (sh)
- C shell (csh)

Distributions of Linux

- Ubuntu
- Debian
- Linux Mint
- Red Hat Enterprise Linux
- Cent OS
- Fedora
- KALI Linux
- Arch Linux
- OpenSUSE